PXR-Challenge, 3D Structure Track Report

Computational Pipeline: Chai-1 Co-folding → GNINA Rigid Docking → Smart Pose Selector & PDB Sanitizer

1. Leaderboard Results

The following metrics represent the validation scores returned by the OpenADME server for the latest successful submission.

Metric	Score	Notes / Status
LDDT-PLI	**0.4285**	Protein–ligand interaction quality (Significant improvement)
BiSyRMSD	4.5709	Bidirectional symmetry-aware RMSD
LDDT-LP	*0.8723*	Ligand pose local distance difference

2. Methodology

The computational pipeline was divided into four distinct phases, combining the predictive power of Deep Learning for the receptor with the geometric precision of classic force fields and Convolutional Neural Networks (CNN) for the ligands.

```
 [FASTA Sequence]
       |
       ▼ (Phase 1: Chai-1 AI Co-folding)

[10 Generated Complexes] ──▶ Best Selection ──▶ Ligand Removal ──▶ [receptor_PXR.pdb]
                                                                         |
       ┌─────────────────────────────────────────────────────────────────┘
       ▼ (Phase 2: GNINA Molecular Docking on Linux)

[184 Docked Fragments (.sdf)]
       |
       ▼ (Phase 3 & 4: Python Post-Processing & PDB Sanitization)

 ├── 182 Molecules ──▶ Top 1 Pose Extraction
 ├── 2 Molecules ──▶ Top 2 Pose Extraction (Steric Clash Resolution)
 └── PDB Formatting (TER addition, Unique IDs, Columns 77-80) ──▶ [Final Submission]
```

Phase 1: Receptor Generation via Chai-1 (AI Co-folding)

To generate the 3D structure of the target PXR:

- The starting point was FASTA sequence of the receptor.

- The sequence was processed using Chai-1 model, generating 10 alternative 3D complexes.
- The 10 models were visually and energetically inspected to select the most stable and structurally sound receptor conformation.
- The native prediction ligand was removed from the selected complex, isolating the clean protein structure.

Phase 2: Virtual Screening and Docking with GNINA

Using the receptor obtained from Chai-1, molecular docking was conducted on all 184 fragments of the challenge.

- **Software:** GNINA (in a Linux environment).
- **Parameters:** Rigid docking (receptor side chains and backbone locked), scoring guided by the integrated convolutional neural network (CNN rescore)

Phase 3: Smart Pose Selection (Clash Resolution)

To prevent server scoring failures caused by physically impossible atomic clashes, a 3D scan for steric clashes was performed (clash threshold set to < 2.0 Å).

- Standard Setup (182 molecules): The highest-scoring pose was extracted and processed. The pocket geometry generated by Chai-1 accommodated these molecules without creating problems.
- Critical Cases (2 molecules): For compounds `x02828-1` and `x03421-1`, the Top 1 pose presented a severe steric clash against the pocket walls, leading to a server crash (*Scoring Failed*). The Top 1 pose was discarded in favor of the Top 2 pose (the second-best geometric conformation generated by GNINA). The Top 2 pose successfully resolved the steric hindrance.

Phase 4: PDB Standard Formatting and Sanitization

The final submission files were reconstructed by merging the atomic coordinates of the receptor and the respective ligand. Formal sanitization was applied via Python scripts:

1. Unique Atom Renumbering: OpenBabel assigns generic names to ligand atoms converted from SDF. The script dynamically renamed each atom uniquely, eliminating structural ambiguity within the same residue.
2. TER Record Insertion: The end of the protein chain (Chain A) was dynamically calculated, and the `TER` record was inserted.
3. Receptor Integrity Restoration: The ATOM lines of the original Chai-1 receptor lacked the final standard PDB columns. The script reconstructed every line.

3. Limitations

1. **Receptor Flexibility (Induced Fit):** Although Chai-1 excellently predicted the starting point, the subsequent GNINA docking was conducted keeping the protein completely rigid. Allowing rotameric movement of key side chains in the pocket could further increase the interaction score.
2. **Local Energy Optimization:** A local relaxation (*Local Minimization* or *Energy Minimization*) using classic force fields would refine hydrogen bond distances, potentially pushing the LDDT-PLI value beyond the current 0.4277.